

Rules

- $0 \leq P(A) \leq 1$
- $P(\phi) = 0$
- $P(S) = 1$
- $P(A^c) = P(S) - P(A) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A - B) = P(A) - P(A \cap B) = P(A \cap B^c)$
- $P(B - A) = P(B) - P(A \cap B) = P(A^c \cap B)$
- $P(B^c \cap A^c) = P(A \cup B)^c = 1 - P(A \cup B)$
- $P(B^c \cup A^c) = P(A \cap B)^c = 1 - P(A \cap B)$
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

If $A \subset B$

$$P(A \cap B) = P(A), \quad P(A \cup B) = P(B)$$

$$\text{If } A \cap B = \emptyset \quad \Rightarrow \quad P(A \cap B) = 0$$

$$P(A - B) = P(A)$$

$$P(B - A) = P(B)$$

$$P(B \cup A) = P(A) + P(B)$$

$$P(A \setminus B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) > 0$$

- $P(\emptyset \setminus B) = 0$
- $P(A^c \setminus B) = 1 - P(A \setminus B)$
- $P(A \setminus B) = P(A \cap C \setminus B) + P(A \cap C^c \setminus B)$
- $P(A \cup C \setminus B) = P(A \setminus B) + P(C \setminus B) - P(A \cap C \setminus B)$

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1. **samplemean** : $\bar{X} = \frac{\sum X}{n}$; **samplevariance** : $S^2 = \frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1}$.
 2. **Binomial** (n, p) : $P(X = k) = {}^nC_k \cdot p^k \cdot (1-p)^{n-k}$, $E(X) = np$, $V(X) = np(1-p)$.
 3. **Geometric** p : $P(X = k) = p \cdot (1-p)^{k-1}$, $E(X) = 1/p$, $V(X) = (1-p)/p^2$.
 4. **Poisson**: $P(X_\lambda = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ $E(X_\lambda) = \lambda = V(X_\lambda)$.
 5. **Exponential**: $P(T \leq t) = F(t) = 1 - e^{-\lambda t}$ $E(T) = 1/\lambda$ & $V(T) = 1/\lambda^2$.
 6. **Standard normal distribution**: is a N.D. with $\mu = 0$ and $\sigma = 1$. $Z = \frac{X-\mu}{\sigma}$.
 7. **Sampling & Central L. T.**: “As the sample size **n** increases, the shape of the distribution of sample means from a population “normal or $n \geq 30$ ” with μ and σ will approach a N.D. with $\mu_{\bar{X}} = \mu$ and $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.” And we use $z = \frac{\bar{X} - \mu}{\sigma} \cdot \sqrt{n}$.
 8. **Binomial $\approx$ N.D.**: If a binomial probability distribution “discrete” satisfies that $n \cdot p \geq 5$ and $n \cdot q \geq 5$, then it can be approximated by a N.D. “continuous” with mean $\mu = n \cdot p$ and a standard deviation $\sigma = \sqrt{npq}$.
 9. **Confidence interval for μ** : If σ is given
 - **2-sided**: $\bar{x} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$
 - **1-sided**: $\mu > \bar{x} - z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$ “= **lower bound**” & $\mu < \bar{x} + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$ “= **upper**”
 10. **Confidence interval for σ^2** :
 - **2-sided**: $\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}}^2}$
 - **1-sided**: $\sigma^2 > \frac{(n-1)s^2}{\chi_\alpha^2}$ “= **lower bound**” & $\sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha}^2}$ “= **upper**”
 11. **Hypothesis Testing**: **p-value** = $P(|Z| > |z_0|)$, where z_0 is the test value
 - (i) For 2-tailed test: $p\text{-value} = 2[1 - \Phi(|z_0|)]$
 - (ii) For left-tailed: $p\text{-value} = \Phi(z_0)$. For right-tailed: $p\text{-value} = 1 - \Phi(z_0)$
 12. The **test power** is “ $1-\beta$ ”, where $\beta = \Phi(z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}) - \Phi(-z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma})$ for **2-tailed** test; as well as a sample size that guarantees this test power is $n \approx \frac{(z_\beta + z_{\alpha/2})^2 \sigma^2}{\delta^2}$.
For 1-tailed: if $\delta > 0$, $\beta = \Phi(z_\alpha - \frac{\delta\sqrt{n}}{\sigma})$. If $\delta < 0$, $\beta = 1 - \Phi(-z_\alpha - \frac{\delta\sqrt{n}}{\sigma})$.